

Question Bank**Unit -1 Fundamentals of Computer Organization & Digital Electronics**

1. Explain basic architecture of computer with suitable diagram?
2. Difference between Microprocessor and microcontroller.
3. Write a note on binary fixed point representation.
4. Draw and explain instruction cycle state diagram?
5. Write short note on Micro, Mini, Mainframe and Super Computer?
6. Convert 1A716 into the decimal number system.
7. Convert given binary number 110010 into octal number system.
8. Draw a block diagram of basic components of a computer system. Explain each component in detail.
9. Explain Interrupt in detail.
10. Convert following Decimal no. to Binary, Oct and hex a) 378 b) 1346.
11. Explain Instruction cycle in detail.
12. With an example write the steps to subtract a large number from a smaller number using 2's complement method.

Unit 2: Communication Interface

1. Draw an Interface diagram of 8KB of EPROM and 8KB of RAM with 8086.
2. Design interface diagram for two chips of 16K x 8 ROM and two chips of 32K x 8 RAM with 8086. Select the EPROM address suitably. The RAM address must start at 00000h
3. What is ADC and DAC? Define General algorithm for ADC interfacing.
4. Explain various modes of operation of 8255 PPI .
5. Differentiate between I/O mapped I/O and memory mapped I/O
6. Explain 8086 Memory Banking.
7. Difference between Memory mapped I/O and I/O mapped I/O
8. Explain peripherals and interfacing with 8086.
9. It is required to interface two chips of 16K X 8 ROM and two chips of 32K X 8 RAM with 8086. Select the EPROM address suitably. The RAM address must start at 00000H
10. Explain in detail the I/O interfacing.
11. List the feature of 8255.